UNIVERSITY PARTNER

UNIVERSITY OF WOLVERHAMPTON

HERALD COLLEGE KATHMANDU

4MM013 - Computational Mathematics

Mathematics Assignment-2

Full Marks: 20

University ID : 2329860

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1. Using Cramer's rule obtain the solutions to the following set of equations:

$$2\,x_1 + x_2 - x_3 = 0$$

$$x_1 + x_3 = 4$$

$$x_1 + x_2 + x_3 = 0$$

(4)

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1.
$$2x_1 + x_2 - x_3 = 0$$
$$x_1 + x_3 = 4$$
$$x_1 + x_2 + x_3 = 0$$

In matrix form,

$$\begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix}$$

Now

$$|A| = 2(0\times1 - 1\times1) - 1(1\times1 - 1\times1) + (-1)(1\times1 - 0\times1)$$
$$= 2(-1) + (-1)$$
$$= -2 - 1$$
$$= -3$$

$$|A_1| = \begin{vmatrix} 0 & 1 & -1 \\ 4 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix}$$
$$= 0 - 1\times(4\times1 - 1\times0) + (-1)(4\times1 - 0\times0)$$
$$= 0 - 4 - 4$$
$$= -8$$

$$|A_2| = \begin{vmatrix} 2 & 0 & -1 \\ 1 & 4 & 1 \\ 1 & 0 & 1 \end{vmatrix}$$
$$= 2(4\times1 - 1\times0) - 0 + (-1)(1\times0 - 4\times1)$$
$$= 8 - 1\cdot(-4)$$
$$= 12$$

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$$|A_3| = \begin{vmatrix} 2 & 1 & 0 \\ 1 & 0 & 4 \\ 1 & 1 & 0 \end{vmatrix}$$

$$= 2(0\times0 - 4\times1) - 1(1\times0 - 4\times1)$$

$$= 2(-4) - 1(-4)$$

$$= -8 + 4$$

$$= -4$$

$$x_1 = \frac{|A_1|}{|A|} = \frac{-8}{-3} = \frac{8}{3}$$

$$x_2 = \frac{|A_2|}{|A|} = \frac{12}{-3} = -4$$

$$x_3 = \frac{|A_3|}{|A|} = \frac{-4}{-3} = \frac{4}{3}$$

$$\therefore\ x_1 = \frac{8}{3},\ x_2 = -4,\ x_3 = \frac{4}{3}$$ Ans

2.

a) Solve the following using Gauss elimination:

$$x_1 + x_2 + x_3 = 2$$

$$2x_1 + 3x_2 + 4x_3 = 3$$

$$x_1 - 2x_2 - x_3 = 1$$

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2. a) Given,

$$x_1 + x_2 + x_3 = 2 \quad \text{---(i)}$$
$$2x_1 + 3x_2 + 4x_3 = 3 \quad \text{---(ii)}$$
$$x_1 - 2x_2 - x_3 = 1 \quad \text{---(iii)}$$

In matrix form,

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 2 \\ 2 & 3 & 4 & 3 \\ 1 & -2 & -1 & 1 \end{array}\right]$$

$R_2 \to R_2 - 2R_1$

$R_3 \to R_3 - R_1$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 2 \\ 2 & 3 & 4 & 3 \\ 1 & -2 & -1 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc:c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & -3 & 2 & 1 \end{array}\right]$$

$R_3 \to R_3 + 3R_2$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 4 & -4 \end{array}\right]$$

$$x_1 + x_2 + x_3 = 2 \quad \text{---(iv)}$$
$$x_2 + 2x_3 = -1 \quad \text{---(v)}$$
$$4x_3 = -4 \quad \text{---(vi)}$$

Substituting the value of x_3 in eqn (v)

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$x_2 + (-1) \cdot 2 = -1$

$x_2 = 1$

Substituting the value of x_2 & x_3 in eqn (iv)

$x_1 + 1 - 1 = 2$

$\therefore x_1 = 2$

Therefore, $x_1 = 2$, $x_2 = 1$, $x_3 = -1$ Ans

b) Find the inverse of the matrix from (a) using elimination.

(4)

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2. b) Here,

$$x_1 + x_2 + x_3 = 2 \quad \text{---} (i)$$
$$2x_1 + 3x_2 + 4x_3 = 3 \quad \text{---} (ii)$$
$$x_1 - 2x_2 - x_3 = 1 \quad \text{---} (iii)$$

In matrix form,

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 1 & -2 & -1 \end{bmatrix}$$

So,
$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 2 & 3 & 4 & 0 & 1 & 0 \\ 1 & -2 & -1 & 0 & 0 & 1 \end{array}\right]$$

$R_2 \rightarrow R_2 - 2R_1$
$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -2 & 1 & 0 \\ 1 & -2 & 1 & 0 & 0 & 1 \end{array}\right]$$

$R_3 \rightarrow R_3 - R_1$
$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -2 & 1 & 0 \\ 0 & -3 & -2 & -1 & 0 & 1 \end{array}\right]$$

$R_3 \rightarrow R_3 + 3R_2$

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$$\left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -2 & 1 & 0 \\ 0 & 0 & 4 & -7 & 3 & 1 \end{array}\right]$$

$R_3 \to R_3/4$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -2 & 1 & 0 \\ 0 & 0 & 1 & -7/4 & 3/4 & 1/4 \end{array}\right]$$

$R_2 \to R_2 - 2R_3$

$R_1 \to R_1 - R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 11/4 & -3/4 & -1/4 \\ 0 & 1 & 0 & 3/2 & -1/2 & -1/2 \\ 0 & 0 & 1 & -7/4 & 3/4 & 1/4 \end{array}\right]$$

$R_1 \to R_1 - R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5/4 & -1/4 & 1/4 \\ 0 & 1 & 0 & 3/2 & -1/2 & -1/2 \\ 0 & 0 & 1 & -7/4 & 3/4 & 1/4 \end{array}\right]$$

$$\therefore A^{-1} = \begin{bmatrix} 5/4 & -1/4 & 1/4 \\ 3/2 & -1/2 & -1/2 \\ -7/4 & 3/4 & 1/4 \end{bmatrix}$$

3. Determine whether the following sequence converges or diverges.

$$t_n = (-1)^{n+1} \frac{n+1}{n^2+3}$$

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3. Here,

$$t_n = (-1)^{n+1} \frac{n+1}{n^2+3}$$

lim on t_n,

$$= \lim_{n \to \infty} (-1)^{n+1} \frac{n+1}{n^2+3}$$

$$= \lim_{n \to \infty} \frac{n+1}{n^2+3}$$

$$= \lim_{n \to \infty} \frac{n^2(1/n + 1/n^2)}{n^2(1 + 3/n^2)}$$

$$= \frac{1/\infty + 1/\infty}{1 + 3/\infty}$$

$$= \frac{0}{1}$$

$$= 0$$

Here, as 0 is the real number. So, the converges.

4. Find the Maclaurin series expansion of **Sinx,** also calculate the radius of convergence. (4).

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4. Here,

$f(x) = \sin x \qquad f(0) = \sin 0 = 0$

$f'(x) = \cos x \qquad f'(0) = \cos 0 = 1$

$f''(x) = -\sin x \qquad f''(0) = -\sin 0 = 0$

$f'''(x) = -\cos x \qquad f'''(0) = -\cos 0 = -1$

$f''''(x) = \sin x \qquad f''''(0) = \sin 0 = 0$

Now,

Maclaurin Series,

$$= f(0) + x\frac{f'(0)}{1!} + x^2\frac{f''(0)}{2!} + x^3\frac{f'''(0)}{3!} + x^4\frac{f''''(0)}{4!}$$

$$= 0 + x + 0 - x^2 + 0$$

$$= x - \frac{x^3}{3!}$$

The general term of given series is,

$$(-1)^{n+1}\frac{x^{2n-1}}{(2^n-1)!}$$

$$a = \frac{1}{(2n-1)!} \qquad a_{n+1} = \frac{1}{(2n-1+1)!} = \frac{1}{2n!}$$

Radius of convergence,

$$= \lim_{n \to \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

$$= \lim_{n \to \infty} \frac{1/(2n-1)!}{1/2n!}$$

$$= \lim_{n \to \infty} \frac{2n!}{(2n-1)!}$$

$$= \lim_{n \to \infty} \frac{2n(2n-1)!}{(2n-1)!}$$

$$= 2 \times \infty$$

$$= \infty$$

The End